

DEBAYAN DAS

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#) • [LeetCode](#)

EDUCATION

Jadavpur University – B.E., Electronics & Tele-Communication Engineering 2023 – 2027
CGPA: 8.2 / 10
• **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management System, Operating Systems, Computer Networks

Siliguri Boys' High School – WBCHSE (PCMB), Class XII 2021 – 2023
Percentage: 90%

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, JavaScript (ES6+), HTML5, CSS3, SQL.
Frameworks: React.js, Node.js, Express.js, RESTful APIs, Responsive UI Design
Databases: MongoDB, MySQL, Neon Postgres.
Tools: Git, GitHub, VS Code, Linux

PROFESSIONAL EXPERIENCE

AlgoBengal Research Labs · *Cryptography Research Intern* · June 2026 – Present
Kolkata, India (On-site)

- Architected a pure-Rust port of the CRYSTALS-Dilithium post-quantum signature scheme from C, enforcing zero unsafe code to guarantee complete memory safety in core systems.
- Proved byte-for-byte functional equivalence against the C reference implementation by engineering and validating bi-directional FFI interoperability.
- Actively participated in technical design discussions and incorporated code review feedback from cross-functional researchers to ensure high-quality, production-ready implementation with minimal defects.

PROJECTS

SpeedAI – AI SaaS Platform [\[Live ↗\]](#)[\[Code ↗\]](#) | *React, Node.js, Express, Neon Postgres*

- Developed a modern AI SaaS web application providing users with AI image generation and article writing, operating seamlessly across a highly responsive frontend built with React 19 and Tailwind CSS.
- Engineered robust backend services using Node.js and Express to securely handle high-volume HTTP requests, bridging the frontend with a Neon Postgres database for reliable state management.
- Integrated Clerk for secure cross-platform authentication and Cloudinary for optimized media management.

Heuristic Search Based Rubik's Cube Solver | *C++, Bitmasking, IDA**

- Designed a high-performance 3x3 Rubik's Cube solver by modeling the cube as a Bitboard, utilizing advanced OOP principles and bitmasking to enable efficient state transitions.
- Implemented the IDA* (Iterative Deepening A*) algorithm, guaranteeing optimal solutions within 15 moves with an average solve time of under 10 seconds.
- Engineered a custom heuristic database using BFS and a permutation indexer to significantly accelerate search depth and performance.

High-Performance Cache Simulator [\[Code ↗\]](#) | *C++, Multi-threading, STL*

- Engineered a thread-safe concurrent cache engine in C++ implementing LRU and LFU eviction policies, using `std::mutex` and templates to ensure safe parallel read/write operations.
- Authored well-documented, high-quality code mimicking production standards to benchmark over 200,000 parallel operations, modeling an 80/20 data locality pattern..
- Dynamically logged hit/miss ratios, eviction rates, and latency metrics to rigorously test algorithm efficiency and ensure project completion with zero concurrency defects.

ACHIEVEMENTS

- Solved **500+ DSA problems** across LeetCode, Codeforces, and other competitive programming platforms.
- Achieved **Knight rating (Highest Rating: 1993)** on [LeetCode](#), placing in the top competitive programming tier.
- Ranked **#198 globally** in LeetCode Weekly Contest 500, **#394 globally** in LeetCode Weekly Contest 507.
- Selected as a **Siemens Scholar** from over 300 candidates across India, recognizing academic and technical excellence.